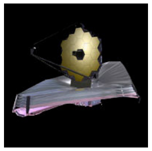

8812 - Distribution of water in the solar system from crossover region spectroscopy of asteroids

Cycle: 4, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

For small bodies, spectral flux is dominated by reflected sunlight in the near-infrared and by thermal emission in the mid-infrared. The crossover region (4-8 m) has contributions from both and their relative contribution to total flux varies based on the asteroid's distance from the Sun. The crossover region is often excluded from analysis of asteroid spectra, likely because the signal to noise at short wavelengths is lower than the rest of the spectrum. Despite this exclusion, the crossover region is home to spectral features indicative of olivine, pyroxene, and molecular water. We propose to use archival MIRI MRS data of 32 small bodies to explore the crossover region. The targets span a range of semimajor axes, inclinations, geometric albedos, rotational periods, and taxonomies, making it a representative sample to understand the distribution of hydration in the solar system. The study of the distribution and evolution of hydrated small bodies has direct implications for how water was delivered to Earth and how water could be delivered to other extrasolar planets.

OBSERVING DESCRIPTION